Pond preparation, stocking and water quality management

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POND PREPARATIONS

- Remove excess silt from the pond bottom (when necessary)
- Dry the pond completely to kill off any remaining fish, and decrease diseases.
- Remove old compost and repair the crib.
- Repair the dikes and rebuild slopes (if necessary)

POND PREPARATIONS

- Cut any grass in or around the pond
- Check on the pipe condition and replace broken or worn out pipes.
- Seal up the pond, that is, close the dike if it was cut during harvest or close completely the drainage pipes.
- Add compost or manure and then put 30 cm of water in the pond for easy blooming. Water should be green before stocking the pond with fish.

Pond Liming

Purpose

- ✓ Controlling acidity of the soil.
- ✓ Disinfecting the pond

Liming rates

Recommended liming rates used in the fish ponds are:

- ➢ Agriculture lime - 3-5kg/100m^2/week
- ➢ Quick lime – 2-3kg/100m^2/week
- ➢ Hydrated lime – 1-2kg/100m^2/week

Pond Fertilization

- Manure application must be started while the ponds are in the process of filling to have abundant food available in the pond when the fingerlings are stocked.
- Once the colour of water in the pond turns green, the pond is ready for stocking.

Pond Fertilization

Types of manure

- **Compost manure** – compost supplies nutrient requirements for growth of phytoplankton which provide major natural food source (10kg of animal manure/100m^2 pond size)
- **Animal manure** – these are basically droppings from livestock such as cows, goats, sheep, chickens, pigs, rabbits and pigeons.
- **Green manure** – these are plant leaves such as Beans, grass, groundnuts, sweet potatoes, sugarcane, banana leaves and stems, sunhemp and cassava

Manure type	Quantity/100m^2	Frequency	Remarks
Pig manure	5 kg	Weekly	**It might not be necessary to continue application of manure if the bloom status is okay to avoid over-fertilization**
Goat manure	5 kg	Weekly	
Cow manure	5 kg	Weekly	
Chicken manure	5 kg	Weekly	
Rabbit manure	5 kg	Weekly	
Green manure	10% of pond area	Weekly	**Fill up the crib and stir daily**
Compost manure	**10% of pond area**	**Weekly**	

FISH HANDLING, TRANSPORTATION AND STOCKING

- In the day-to-day operations of a fish farming enterprise, one cannot avoid handling fish
- Any form of handling causes stress in the fish and could potentially lead death if prolonged
- Do not touch a fish with dry hands.
- Always have in mind a way to get fresh water to fish, especially when sorting during harvest.
- When fish are near the water surface, they are calling for help.
- Use the proper mesh size for fish if hand-netting or seining.
- Fish should be held in the cleanest water possible.

FISH HANDLING, TRANSPORTATION AND STOCKING



- Use appropriate mode of transport which will reduce stress
- Plan to do the transport early in the morning before it gets too hot.

POND STOCKING

To stock a pond, the following things must be clearly understood;

1. Type of fish to be stocked (mixed or monosex fish)

2. The size of a pond

3. Types of management

For business, farmers are encouraged to use monosex fish

POND STOCKING

- For monosex fish, atleast five fish can be stocked per square meter (m2)
- Depending on the management, some time the number offish can be to be stocked can be increased (if aeration is to be provided)
- For mix sex fish the number of fish to be stocked reduced a bit.
- This is done to avoid overstocking the pond or to follow the stocking density.

POND STOCKING

- The fingerlings must not be thrown in the pond, but must be acclimatised and released slow/allowed to swim out on their own
- This reduced the mortality due the shock in the variability of environments

END OF PART ONE

PRODUCTION OF MONOSEX

In aquaculture, monosex populations desired due variations that come with the differences in sex for economic reasons.

Fish provides a better opportunity to manipulate fertilized eggs/fry than in mammals or birds because;

1. Fertilization in fish occurs externally, and
2. Embryological development takes place without the protection of an internal womb or a hard-shelled egg.

PRODUCTION OF MONOSEX

3.During early embryology, an embryo is, phenotypically, neither male nor female in that it does not possess ovaries, testes or other characteristics associated with the reproductive systems.

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

- The problem with Tilapia culture is uncontrolled reproduction that may occur in grow-out ponds.
- Monosex culture is one of the basic methods of controlling Tilapia populations that have been carried out in some countries for aquaculture purposes.

Some methods of monosex tilapia production: A review

Eze Felix[1*], FE Avwemoya[2], Abel Abah[3]

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[2] Department of Fisheries and Aquaculture, Faculty of Marine Environmental Management, Nigeria Maritime University, Okerenkoko, Delta State, Nigeria

[3]Department of Aquaculture and Fisheries Management, Faculty of Renewable Natural Resources, University of Ibadan, Oyo State, Nigeria

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

This technique includes;

1. Manual separation of sexes
2. Environmental manipulation
3. Hybridization
4. Hormone augmentation (sex reversal) and
5. Genetic manipulation methods such as androgenesis, gynogenesis, polyploidy and transgenesis.

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

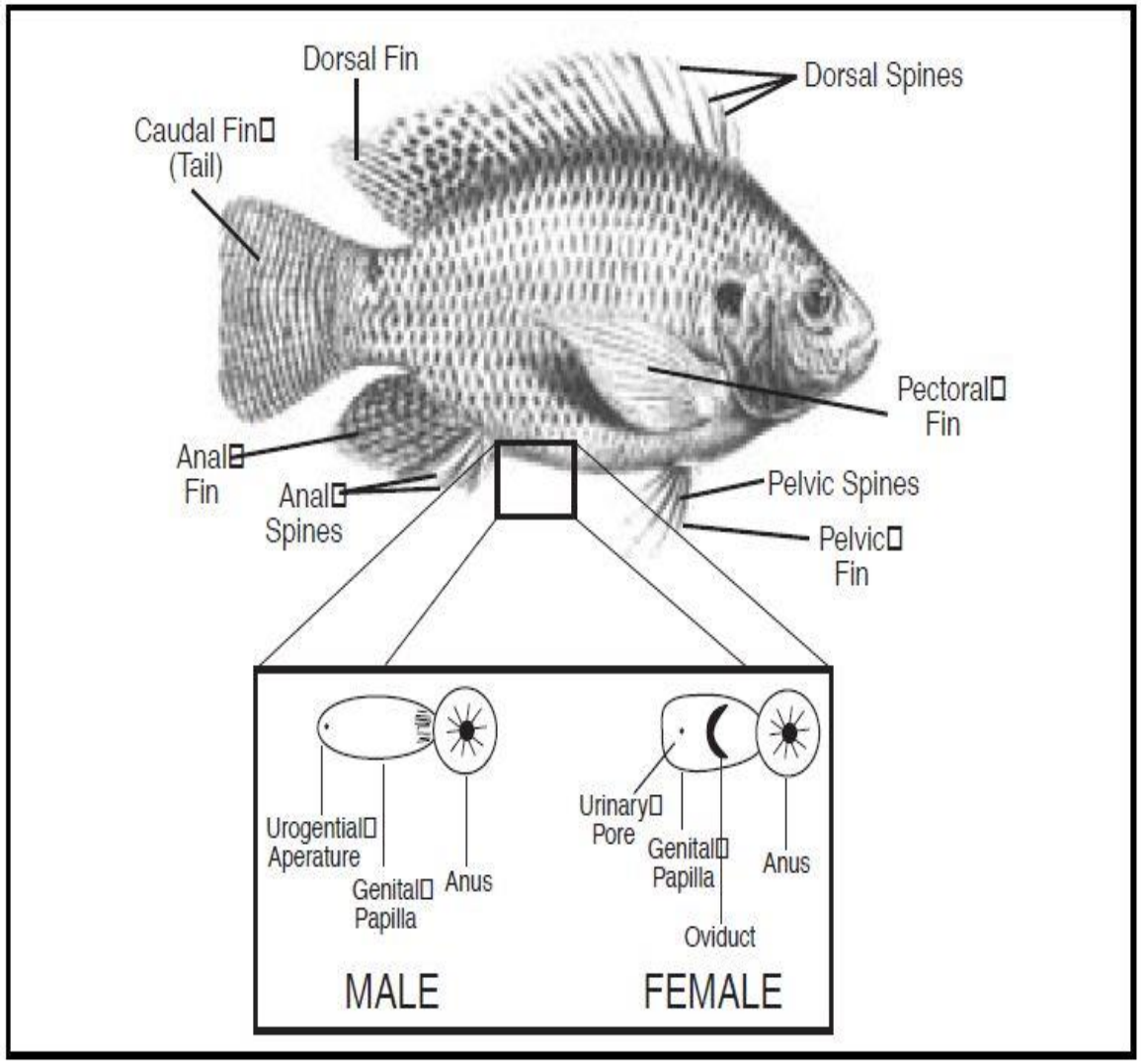
❑ Males are preferred because;

1. They grow almost twice as fast as females, which may be caused by a sex specific physiological growth capacity.
2. Female mouthbrooding.
3. More aggressive feeding behavior of males.
4. Expected survival for all-male culture is 90% or greater.

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

1. Manual Sorting

Manual sorting involves separating males from females by visual inspection of the external urogenital pores.



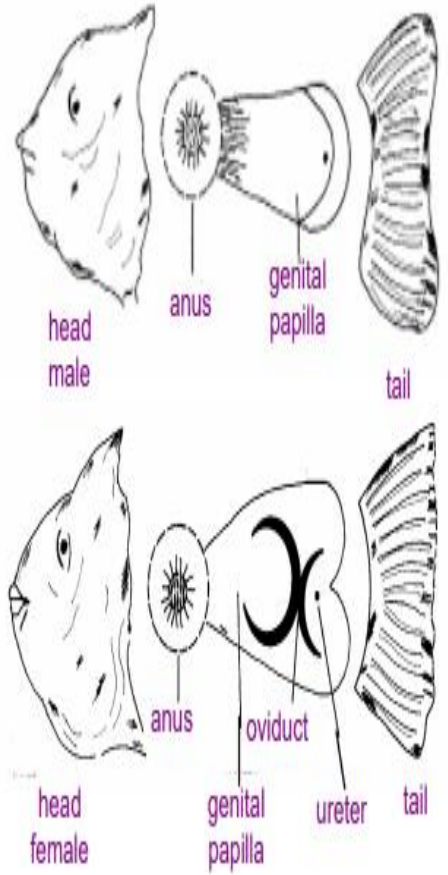


Figure 1. Sex differentiation of male and female tilapia

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

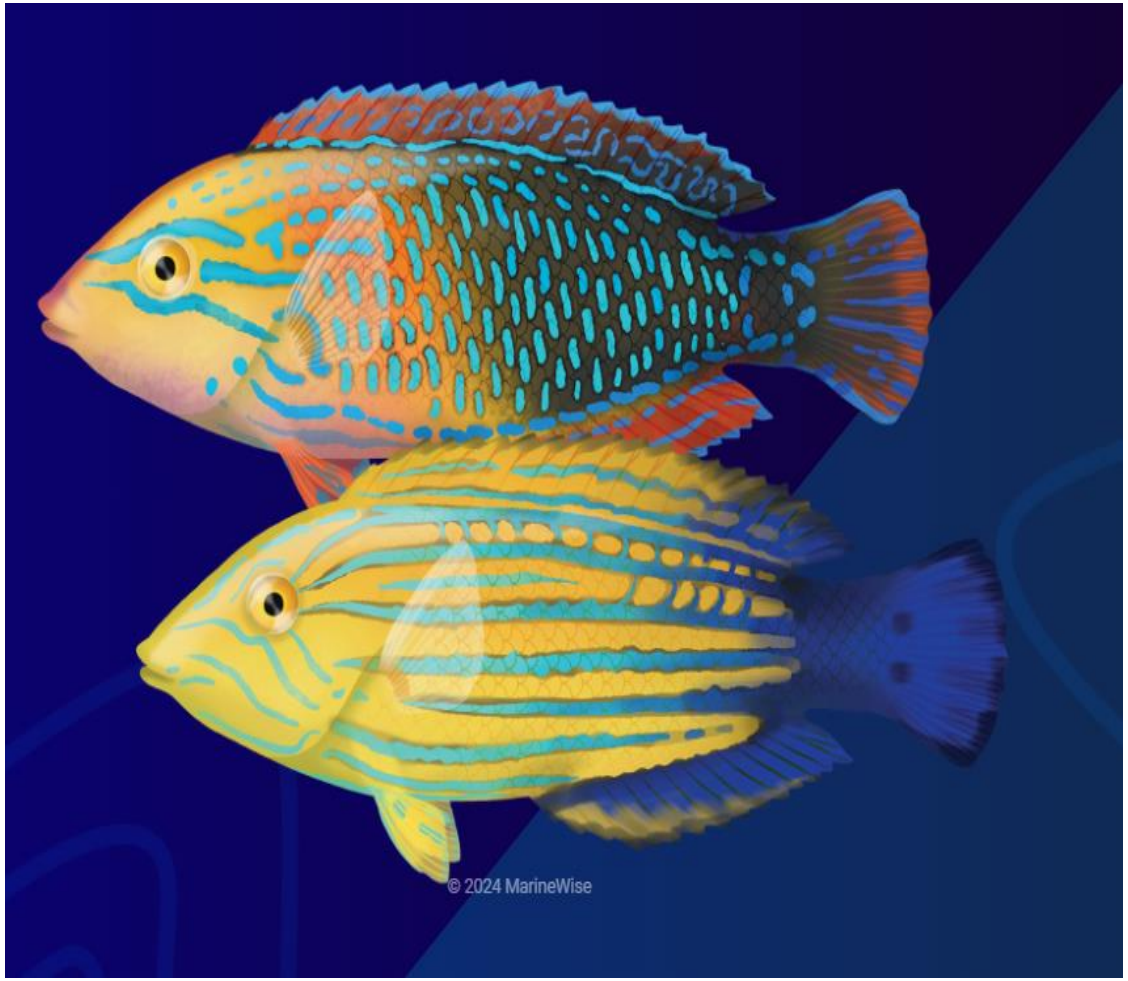
Sexual dimorphism

- Two sexes of the same species exhibit different characteristics apart from the differences in their sexual organs.
- Such differences may be in color, shape and size, which makes males distinct

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

Sexual dimorphism

Bluetail Wrasse exhibit distinct sexual dimorphism. Males are brightly coloured, with a vibrant blue body, yellow head, and black markings on the dorsal and anal fins. Females, on the other hand, have a duller appearance with a Gray-brown body and a yellow tail.



SOME METHODS OF MONOSEX TILAPIA PRODUCTION

- **2.Environmental manipulation**

- Fish show particularities in their temperature sex determination patterns since monosex populations are generally not produced at extreme temperatures, suggesting the existence of strong temperature genotype interactions.

- Temperature treatments must be applied at a critical sensitive period, relatively similar to the hormone sensitive period.

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

- **2.Environmental manipulation**

- The highest masculinity rate (Conversion of females to male) was obtained from fishes reared under 30 and 36°C respectively (73.68% and 86.67%).

Effects of Water Temperature on Sex Differentiation and Growth Parameters of the Mozambique Tilapia (*Oreochromis mossambicus*, Peters, 1852)

Karl Christofer Kingueleoua Koyakomanda, Muamer Kürşat Fırat, Cüneyt Süzer, Serhat Engin, Müge Hekimoğlu, Hülya Sayğı, Osman Özden, Fatih Güleç, Şahin Saka

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SOME METHODS OF MONOSEX TILAPIA PRODUCTION

➢ 3.Hybridization

Hybridization may involve combining different strains of a species (that is members of the same species with different characteristics)

Table 1: Hybridization of some tilapia species and proportion of male progeny produced

Crosses (♂ x ♀)	Males (%)	References
O. aureus x O. niloticus (Ugadan strain)	96-100	Pruginin, 1967
O.hornorum x O. spilurus	100	Hulata *et al.*,, 1983
O. aureus x O. vulcani	98-100	Pruginin, 1975
O. macrochir x O. niloticus	100	Wohlfarth *et al.*,, 1994
O.urolepsis hormorum x O. nigra	98-100	Wohlfarth *et al.*,, 1994
O.urolepsis hormorum x O. vulcani	98-100	Majumdar *et al.*,, 1983
O. macrochir x O. mossambucus	100	Majumdar *et al.*,, 1983
O.hornorum x O. niloticus	100	Wohlfarth *et al.*,, 1994
O.urolepsis hormorum x O. niloticus	100	Wohlfarth *et al.*,, 1990
O. aureus x O. niloticus (Stirling strain)	100	Marengoni *et al.*,, 1998
O. aureus x O. mossambicus	100	Beadmore *et al.*,, 2001
O.hornorum x O. mossambicus	100	Hickling, 1960

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

- **4. Sex Reversal or Hormone Augmentation**

- The principle behind this method lies on the fact that at the stage when the tilapia larvae are said to be sexually undifferentiated (right after hatching up to about 2 weeks or up to the swim-up stage) the extent of the androgen (male hormone) and the estrogen (female hormone) present in a fish is equal.

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

- **4. Sex Reversal or Hormone Augmentation**

- Thus, increasing one of the hormones that is originally present in the fish will direct the fish to either male or female depending upon the hormone introduced.

- Accordingly, if the tilapia larvae are fed with feeds that are incorporated with male hormone (e.g.17α-methyltestosterone), the fish will develop into phenotypic male (physically appears and functions as male but possesses the female genotype (XX).

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

4. Sex Reversal or Hormone Augmentation

- Feeding the larvae with hormonetreated diet, e.g., 17α-methyltestosterone or estrogen between the second and sixth week after hatching has been observed to have produced high percentage of males and females, respectively.

Table 2: List of successful use of different type of hormones in producing monosex tilapia

Species	Hormone	Duration	Male (%)	References
Oreochromis niloticus	Fadrozole	30 days	92.5-96	Kwon *et al.*,, (2000)
O. niloticus	17 α-ethynyltestosterone	25-28 days	91-99.4	Afonso *et al.*, (2001)
O. niloticus	17 α methyldihydrotestosterone	21 days	99	Guerrero and Guerror (1988) [48]
O. niloticus	17 α methyldihydrotestosterone	4 h	100	Wassermann and Afonso (200
O. mossambicus	17 α-ethynyltestosterone	18 days	100	Guerrero (1975) [35]
O. mossambicus	17 α-methyltestosterone	42 days	100	Hines and Watts (1995
O. aureus	17 α-ethynyltestosterone	25-28 days	83-97	Melard *et al.*, (1994)

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

4. Sex Reversal or Hormone Augmentation

- This method can be performed by oral administration of feed incorporated with androgen and eggs or fry immersion in different concentrations of the male hormone.
- Hormones may also be difficult to obtain in some countries and hatchery facilities

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SOME METHODS OF MONOSEX TILAPIA PRODUCTION

4. Sex Reversal or Hormone Augmentation

- Sex reversal by the immersion technique is achieved by immersing the eggs in different concentrations of 17- αmethyl testosterone exposed for different times.



SOME METHODS OF MONOSEX TILAPIA PRODUCTION

4. Sex Reversal or Hormone Augmentation

- The mechanism of action of the immersion technique is that the hormone is absorbed through passive diffusion across the lipid membrane of the eggs.
- During the embryonic development, gonadal differentiation can be affected by the administration of steroid sex hormone in the holding water.

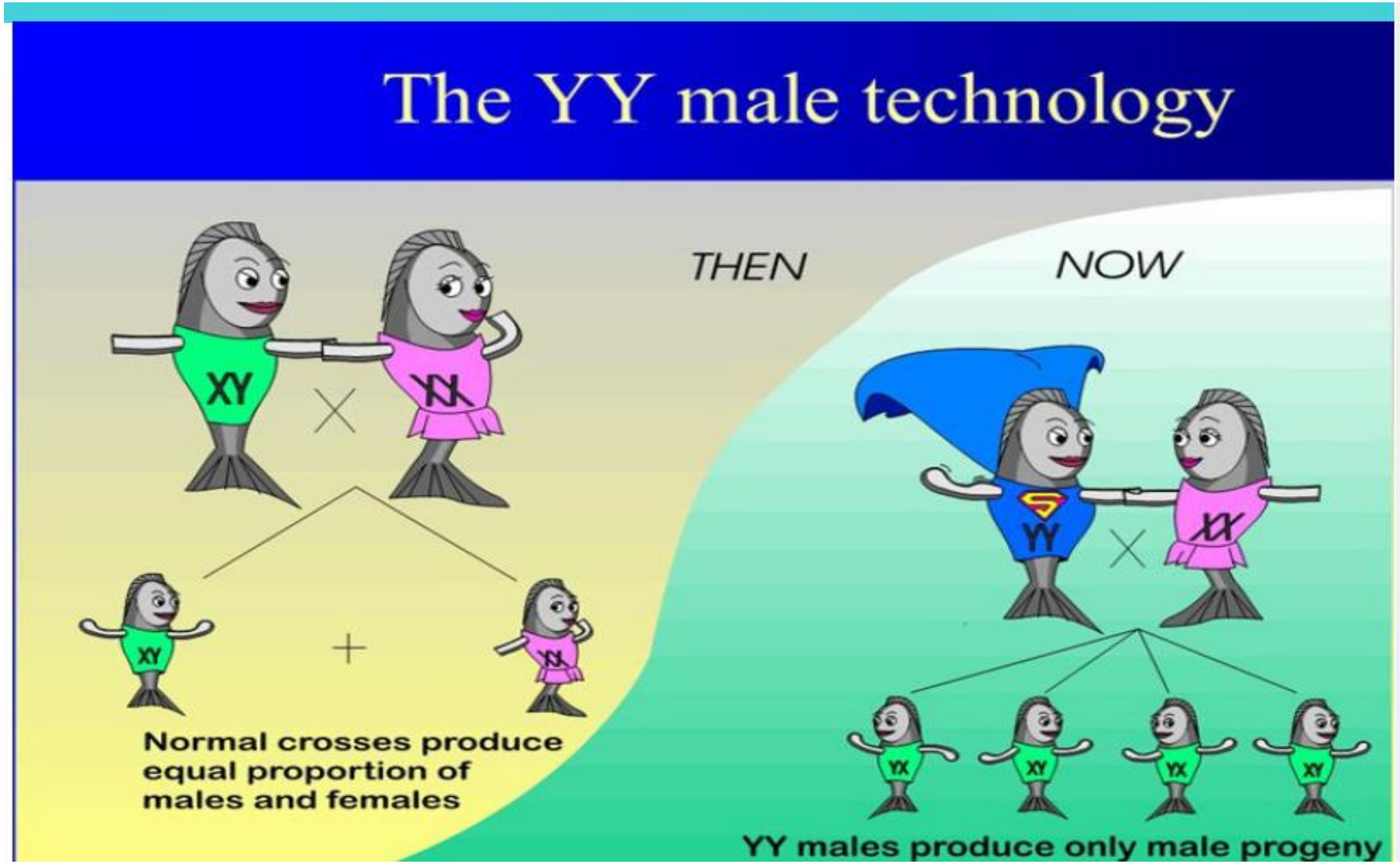
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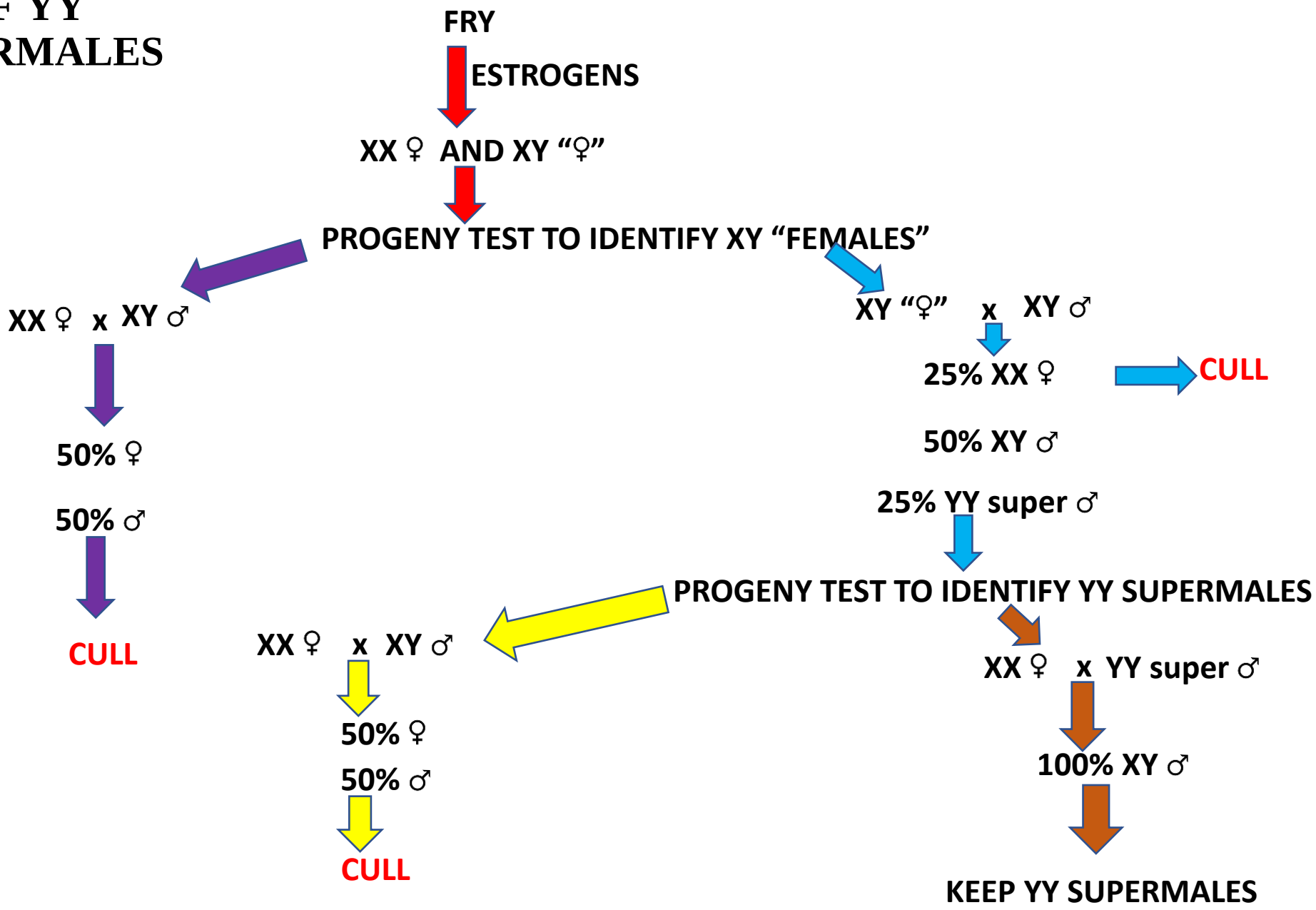
- 800 µg L-1 hormone concentration at 96-h Immersion time gave 91% reversal rate.



SOME METHODS OF MONOSEX TILAPIA PRODUCTION



PRODUCTION OF YY SUPERMALES



SOME METHODS OF MONOSEX TILAPIA PRODUCTION

4. Sex Reversal or Hormone Augmentation

- For sex-reversal, 60 mg of 17 α -methyl-testosterone (MT) is mixed in 15ml 95% ethanol;

- Then the solution is mixed with 1 kg of fish meal (or high protein fry booster mass containing 50% crude protein) together with vitamin C.

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

4. Sex Reversal or Hormone Augmentation

- Feed should not be dried under sunlight because the hormone will degenerate;
- After drying, the feed should be packed in a plastic bag or kept in a container with a tight lid and stored in a room at a low temperature (approx. 10°C);
- You can use a cooler.

SOME METHODS OF MONOSEX TILAPIA PRODUCTION

4. Sex Reversal or Hormone Augmentation

- After Sex Reversal Treatment (SRT) the fingerlings should be counted before stocking them for further nursing;
- Fingerling smaller are susceptible to predation by carnivorous fish and birds;
- Smaller fish are also less tolerant to poor water quality.

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